The partition-representation degrees of the dual Fano matroid and a six-symbol characterization of regularity

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Abstract

For a finite matroid M, let $\chi(M)$ be the set of integers $v \geq 2$ for which M has a partition representation of degree v, equivalently a variable-strength orthogonal array of level v. We prove that the dual Fano matroid satisfies

$$\chi(F_7^*) = \{2^k : k \geq 1\}.$$

The proof fixes one coordinate fibre, identifies the resulting contraction with $M(K_4)$, applies Matúš's group normal form, and propagates that local normal form through the remaining circuit equations. The propagation forces every element of the underlying group to be an involution. Combining this classification with the excluded-minor characterization of regular matroids gives a fixed-alphabet characterization: a finite matroid is 6-entropic if and only if it is regular. Moreover, among all integers $v \geq 2$, the class of v-entropic matroids coincides with the class of regular matroids if and only if $v = 6$.

1 Introduction

A partition representation of a matroid is a nonlinear analogue of a linear representation. It may be expressed equivalently as an almost-affine code, an ideal secret-sharing representation, or a variable-strength orthogonal array (VOA). If a matroid M has rank function r_M, a degree-v representation realizes the entropy vector

$$A \longmapsto r_M(A) \log v.$$

Following Abbe and Spirkl, we call M v-entropic when its rank function is realized in this way [3]. Following Chen, Cheng, and Bai, we write $\chi(M)$ for the set of all admissible degrees [2].

In the proof of Proposition 4.1, Matúš showed that the Fano matroid has precisely the power-of-two degrees [1]. For the dual Fano matroid, Abbe and Spirkl gave an analytic proof that degree 3 is impossible [3, Lemma 9]; Chen, Cheng, and Bai later obtained an independent finite-computation proof [2, Lemma 4]. The latter authors also posed the general question whether $\chi(M) = \chi(M^*)$ for every matroid [2, Conjecture 1]. Our first result settles the complete degree set for this particular dual pair.

Theorem 1.1. *For the dual Fano matroid,*

$$\chi(F_7^*) = \{2^k : k \geq 1\}.$$

The argument is structural rather than enumerative. A fibre of a hypothetical degree-v representation of F_7^* represents $M(K_4)$. Matúš's classification of partition representations of $M(K_4)$ supplies a finite group of order v. Three further circuit equations then force that group to have exponent two.

The result has a consequence that is not restricted to a seven-element matroid. Recall that a finite matroid is regular if it is representable over every field, equivalently if it has no minor isomorphic to $U_{2,4}$, F_7, or F_7^*.

Theorem 1.2. *A finite matroid is 6-entropic if and only if it is regular.*

Chen, Cheng, and Bai proved that every regular matroid has a partition representation of every degree [2, Proposition 7]. The new direction in Theorem 1.2 follows by excluding the three regular-matroid obstructions at degree six: the obstruction $U_{2,4}$ is Euler's order-six exception for orthogonal Latin squares, while F_7 and F_7^* are excluded by their power-of-two degree sets.

The alphabet size six is characterized by this property.

Corollary 1.3. *For an integer $v \geq 2$, the following are equivalent:*

1. *every v-entropic matroid is regular;*
2. *the v-entropic matroids are exactly the regular matroids;*
3. $v = 6$.

Indeed, F_7 is a nonregular 2-entropic matroid. For every $v \geq 3$, $v \neq 6$, the nonregular matroid $U_{2,4}$ is v-entropic by the complete existence theorem for pairs of orthogonal Latin squares.

The group slice used below is classical. Recent work of Kalantari and Khazaei also uses Matúš's $M(K_4)$ normal form in its characterization of 4-entropic matroids [4]. Our contribution is the propagation across the specific circuit system of F_7^*, the resulting complete degree classification, and its degree-six consequences.

2 Partition representations and circuit equations

Let $M = (E, r)$ be a finite matroid of rank $r(E)$, and let Q be a set of size v. A variable-strength orthogonal array $\mathrm{VOA}(M, v)$ is a $v^{r(E)} \times |E|$ array T over Q such that, for each $A \subseteq E$, every row appearing in the projection $T(A)$ appears exactly $v^{r(E)-r(A)}$ times. Its coordinate level partitions form a partition representation of degree v, and the two notions are equivalent [1, 2]. The entropy characterization of Chen, Cheng, and Bai shows that M is v-entropic exactly when such an array exists [2, Theorem 1 and the discussion following it]; see also [4, Proposition 2.3].

We use three standard consequences.

Lemma 2.1 (basis coordinates). *If B is a basis of M, every tuple in Q^B occurs exactly once in $T(B)$. Hence each coordinate outside B is a function of the basis coordinates.*

Proof. Since $r(B) = |B| = r(E)$, the VOA multiplicity on B is one. There are $v^{r(E)} = |Q^B|$ rows, so the projection is a bijection. □

Lemma 2.2 (circuit quasigroups). *Let C be a circuit of M and $e \in C$. In every degree-v representation, the coordinate X_e is a quasigroup function of the coordinates $X_{C\setminus\{e\}}$.*

Proof. The set $C \setminus \{e\}$ is independent and $r(C) = |C| - 1$. Every tuple on $C \setminus \{e\}$ therefore has a unique extension to the joint support on C. Repeating this with each coordinate singled out gives the quasigroup property. □

Here a k-ary quasigroup is a map $q : Q^k \to Q$ for which fixing any (k-1) arguments makes the remaining one a bijection.

Lemma 2.3 (minor closure). *If $v \in \chi(M)$ and N is a minor of M, then $v \in \chi(N)$.*

Proof. Deletion is obtained by projecting the array and deduplicating rows. Contraction by an element is obtained by fixing one coordinate value and restricting to that fibre. Iterating these two operations gives every minor. This is [2, Theorem 2]. □

Coordinate relabellings preserve a VOA. More precisely, one may independently apply a bijection $Q \to Q$ to each coordinate. In Matúš's terminology this is partition isotopy. The same relabellings transform all circuit equations simultaneously; they cannot be chosen separately for different fibres.

3 A group slice inside F_7^*

We use the labelling of F_7^* from [2]. Its seven circuit-hyperplanes are

$$1235, \qquad 1246, \qquad 1347, \qquad 1567, \qquad 2367, \qquad 2457, \qquad 3456. \tag{3.1}$$

The set (1234) is a basis. By Lemmas 2.1 and 2.2, a degree-v representation can be written in basis coordinates as

$$\begin{aligned} &X_1 = a, && X_2 = b, && X_3 = c, && X_4 = d, \\ &X_5 = F(a,b,c), && X_6 = G(a,b,d), && X_7 = H(a,c,d), \end{aligned} \tag{3.2}$$

where F, G, H are ternary quasigroups on an alphabet Q of size v.

Fix an arbitrary value $a = a_0$. Restricting to this fibre is the VOA contraction by element (1). The circuit-hyperplanes through (1) give the four triangles

$$235, \qquad 246, \qquad 347, \qquad 567. \tag{3.3}$$

They are precisely the four triangles of a labelled copy of $M(K_4)$. For example, a binary representation of F_7^* may be chosen with

$$5 = e_1 + e_2 + e_3, \qquad 6 = e_1 + e_2 + e_4, \qquad 7 = e_1 + e_3 + e_4.$$

After contraction by e_1, the six nonzero columns are

$$e_2, e_3, e_4, e_2 + e_3, e_2 + e_4, e_3 + e_4,$$

so there are no loops, parallel elements, or additional triangles.

Matúš's classification now supplies the key local normal form.

Lemma 3.1 (group normal form)**.** *After fixed relabellings of the six coordinate alphabets* $2, \ldots, 7$*, there is a finite group* $(\Gamma, \cdot)$ *of order* v *such that, on the fibre* $a = a_0$*,*

$$F(a_0, b, c) = b^{-1}c, \qquad G(a_0, b, d) = b^{-1}d, \qquad H(a_0, c, d) = c^{-1}d. \tag{3.4}$$

The group is not assumed commutative.

Proof. Matúš's Proposition 3.1 states that every partition representation of $M(K_4)$ is partition-isotopic to the canonical representation obtained from a finite group of the same order [1]. One orientation of that canonical model is

$$(x, y, z, xy^{-1}, yz^{-1}, zx^{-1}). \tag{3.5}$$

Put $b = x^{-1}$, $c = y^{-1}$, and $d = z^{-1}$. Keep the output coordinates corresponding to xy^{-1} and yz^{-1}, and invert the coordinate corresponding to zx^{-1}. After matching these coordinates with the triangles in (3.3), the three functions become exactly (3.4). These substitutions are six fixed coordinate bijections. We apply them to the entire representation and use the identity relabelling on coordinate 1; no relabelling depends on a. □

4 Propagation through the remaining circuits

The three circuit-hyperplanes $2367, 2457, 3456$ propagate the normalized fibre to every value of a. The point requiring care is that a circuit equation is initially known on the joint support of its four coordinates. In each of the following steps, the normalized fibre covers the complete three-dimensional input domain of the relevant circuit quasigroup.

Lemma 4.1. *There is, for each* $a \in Q$*, a permutation* $T_a : \Gamma \to \Gamma$ *such that*

$$G(a, b, d) = b^{-1}T_a(d), \qquad H(a, c, d) = c^{-1}T_a(d). \tag{4.1}$$

Proof. Orient the circuit 2367 toward coordinate 7. On the normalized fibre, write $g = b^{-1}d$, so $d = bg$. Then

$$h = c^{-1}d = c^{-1}bg.$$

As b, c, g independently range over Γ, this determines the entire ternary circuit quasigroup. Therefore, for every basis tuple,

$$H(a, c, d) = c^{-1}bG(a, b, d). \tag{4.2}$$

The left-hand side is independent of b. Thus, for fixed a, d, the product $bG(a, b, d)$ is independent of b; call it $T_a(d)$. Since G is a quasigroup in d, T_a is a permutation. Equation (4.1) follows. $\square$

Lemma 4.2. *For every $a \in Q$, there is an element $s_a \in \Gamma$ such that*

$$\begin{aligned} F(a, b, c) &= b^{-1} s_a^{-1} c, \\ G(a, b, d) &= b^{-1} s_a d, \\ H(a, c, d) &= c^{-1} s_a d. \end{aligned} \tag{4.3}$$

Proof. Orient 2457 toward coordinate 7. On the normalized fibre, put $f = b^{-1}c$, so $c = bf$. Then

$$h = c^{-1}d = f^{-1}b^{-1}d.$$

The variables b, d, f independently range over Γ, so the whole circuit quasigroup is determined and

$$H(a, c, d) = F(a, b, c)^{-1} b^{-1} d. \tag{4.4}$$

Substituting (4.1) into (4.4) gives

$$F(a, b, c)^{-1} = c^{-1} T_a(d) d^{-1} b.$$

The left-hand side is independent of d. Hence $T_a(d)d^{-1}$ is constant in d; denote it by s_a. Thus $T_a(d) = s_a d$, and (4.3) follows. $\square$

Lemma 4.3. *Every element s_a in (4.3) is an involution.*

Proof. Orient 3456 toward coordinate 6. On the normalized fibre, $f = b^{-1}c$ gives $b = cf^{-1}$, and hence

$$g = b^{-1}d = fc^{-1}d.$$

The variables c, d, f independently range over Γ. Consequently the complete circuit equation is

$$G(a, b, d) = F(a, b, c)c^{-1}d. \tag{4.5}$$

Using (4.3) on the two sides of (4.5) yields

$$b^{-1} s_a d = b^{-1} s_a^{-1} d.$$

Thus $s_a = s_a^{-1}$, or $s_a^2 = e$. $\square$

We can now prove the first main theorem.

Proof of Theorem 1.1. Fix $b, c \in \Gamma$. Since F is a ternary quasigroup, the map

$$a \longmapsto F(a,b,c) = b^{-1}s_a^{-1}c$$

is a bijection. Hence $a \mapsto s_a$ is a bijection from the alphabet Q onto Γ. By Lemma 4.3, every element $x \in \Gamma$ satisfies $x^2 = e$. Therefore

$$xy = (xy)^{-1} = y^{-1}x^{-1} = yx$$

for all $x, y \in \Gamma$. The group is elementary abelian of order 2^k for some $k \geq 1$, proving that every degree is a power of two.

Conversely, F_7^* is binary. A binary representing matrix has the same column ranks over every extension field $\mathbb{F}_{2^k}$, and the associated linear array is a degree-2^k partition representation. Hence every 2^k occurs. □

5 Six-symbol entropy and regularity

We now prove the global consequence.

Proof of Theorem 1.2. Suppose first that M is 6-entropic. By the entropy–VOA–partition equivalence, $6 \in \chi(M)$. Lemma 2.3 implies that every minor of M also has degree six.

The matroid $U_{2,4}$ has a degree-v representation exactly when there is an index-one orthogonal array $\mathrm{OA}(2, 4, v)$, equivalently a pair of orthogonal Latin squares of order v. Such a pair does not exist for $v = 6$; equivalently, $6 \notin \chi(U_{2,4})$ [2, Example 4]. The proof of Matúš's Proposition 4.1 gives $\chi(F_7) = \{2^k : k \geq 1\}$ [1], and Theorem 1.1 gives the same degree set for F_7^*. Thus none of

$$U_{2,4}, \qquad F_7, \qquad F_7^*$$

is a minor of M. Tutte's excluded-minor theorem for regular matroids implies that M is regular [5, Theorem 6.6.6].

Conversely, let M be regular. A totally unimodular representation of M, reduced modulo v, defines a VOA over the ring $\mathbb{Z}_v$ for every integer $v \geq 2$: every basis determinant is ± 1, hence remains a unit. This is proved explicitly in [2, Lemma 3 and Proposition 7]. In particular $6 \in \chi(M)$, so M is 6-entropic. □

Proof of Corollary 1.3. The implication (2)⇒(1) is immediate, and Theorem 1.2 gives (3)⇒(2).

It remains to show that (1) forces $v = 6$. If $v = 2$, the nonregular Fano matroid F_7 is v-entropic. If $v \geq 3$ and $v \neq 6$, there exists a pair of orthogonal Latin squares of order v, and hence the nonregular matroid $U_{2,4}$ is v-entropic. The existence statement is the theorem of Bose, Shrikhande, and Parker together with the classical small orders [2, Example 4]; [6]. Therefore (1) fails for every $v \neq 6$. □

6 Scope and relation to previous work

The proof uses several established ingredients whose roles should be kept separate from the new argument.

1. Matúš introduced partition representations, classified the $M(K_4)$ representations by finite groups, and determined $\chi(F_7)$ [1].
2. Abbe and Spirkl gave an analytic proof that $3 \notin \chi(F_7^*)$ [3, Lemma 9]. Chen, Cheng, and Bai developed the VOA and entropy formulation, proved minor closure and arbitrary-degree representability of regular matroids, and later gave an independent finite-computation proof of the same degree-three obstruction [2].
3. Kalantari and Khazaei recently used the same $M(K_4)$ group slice in their degree-four characterization [4]. Thus the group normal form itself is not claimed as new.
4. Tutte's excluded-minor characterization and the orthogonal-Latin-square existence theorem are classical [5, 6].

The new point is that the three remaining circuit equations of F_7^* make the local group normal form global and force exponent two. This yields the complete degree set of F_7^*, rather than the exclusion of one fixed degree. Theorems 1.2 and Corollary 1.3 are consequences at the level of all finite matroids.

This result verifies $\chi(F_7) = \chi(F_7^*)$ for the Fano pair, but it does not prove the general duality conjecture of [2]. Nor does it classify partition representations up to partition isotopy; it classifies only the possible degrees.

The literature comparison made for this manuscript found no direct statement of Theorem 1.1, Theorem 1.2, or Corollary 1.3 in the sources above. This is a bounded prior-art statement, not a claim of absolute global priority.

7 Computational calibration

The proof is noncomputational. A small verification script accompanies the research record only to detect multiplication-order errors in the propagation formulas. It checks the full F_7^* VOA conditions for elementary abelian 2-groups of orders $2, 4, 8$, and checks that cyclic groups of orders $4, 6$ and the nonabelian group S_3 fail at the final circuit as predicted. Independent audit calculations also used the dihedral and quaternion groups of order eight as noncommutative negative controls.

These finite checks do not establish any universal assertion and are not used in the proof.

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